

# The Finite Element Method in One Dimension

ME/BME 736 – Advanced Finite Element Methods

Prashant K. Jha

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This handout develops the one-dimensional finite element method in detail. The starting point is a variational problem of the type already developed in the preceding course material. The emphasis here is on the new finite element ideas: Galerkin approximation, meshes, local polynomial basis functions, local and global node numbering, construction of linear and higher-order basis functions, the reference element, element matrices and vectors, local-to-global assembly, and numerical quadrature.

The one-dimensional setting is useful because every step of the construction can be written explicitly and visualized. The same structure will later carry over to finite elements in two and three dimensions.

## 1 Starting point: a one-dimensional variational problem

Consider

$$-\frac{d}{dx} \left( a(x) \frac{du}{dx} \right) + c(x)u = f(x), \quad x \in (0, L), \quad (1)$$

with

$$u(0) = 0, \quad a(L)u'(L) = q_L.$$

Assume  $a(x) > 0$  and  $c(x) \geq 0$ . The corresponding variational problem is

$$\boxed{\text{find } u \in \mathcal{V} \text{ such that } B(u, v) = F(v) \quad \forall v \in \mathcal{V},} \quad (2)$$

where

$$\mathcal{V} = \{v \in H^1(0, L) : v(0) = 0\},$$

$$B(u, v) = \int_0^L [a(x)u'(x)v'(x) + c(x)u(x)v(x)] \, dx, \quad (3)$$

$$F(v) = \int_0^L f(x)v(x) \, dx + q_L v(L). \quad (4)$$

The exact solution belongs to the infinite-dimensional space  $\mathcal{V}$ . The goal is to replace this problem by one involving only finitely many unknown coefficients.

**Axial-bar specialization.** For a one-dimensional elastic bar,

$$a(x) = A(x)E(x), \quad c(x) = 0,$$

where  $A$  is the cross-sectional area and  $E$  is Young's modulus. The unknown  $u$  is the axial displacement. If  $P$  is an end load at  $x = L$ , then  $q_L = P$ . Thus

$$B(u, v) = \int_0^L AE u'v' \, dx, \quad F(v) = \int_0^L f v \, dx + Pv(L).$$

This bar problem will be used repeatedly below.

## 2 Galerkin approximation before finite elements

Let  $\mathcal{V}_N \subset \mathcal{V}$  be an  $N$ -dimensional space spanned by known basis functions  $\psi_1, \dots, \psi_N$ :

$$\mathcal{V}_N = \text{span}\{\psi_1, \dots, \psi_N\}.$$

The Galerkin approximation seeks

$$u_N(x) = \sum_{j=1}^N U_j \psi_j(x)$$

such that

$$B(u_N, v_N) = F(v_N) \quad \forall v_N \in \mathcal{V}_N. \quad (5)$$

Writing

$$v_N(x) = \sum_{i=1}^N V_i \psi_i(x)$$

and using bilinearity of  $B$  and linearity of  $F$  gives

$$\sum_{i=1}^N V_i \left[ \sum_{j=1}^N B(\psi_j, \psi_i) U_j - F(\psi_i) \right] = 0.$$

Because the coefficients  $V_i$  are arbitrary,

$$\sum_{j=1}^N B(\psi_j, \psi_i) U_j = F(\psi_i), \quad i = 1, \dots, N.$$

Define

$$K_{ij} = B(\psi_j, \psi_i), \quad F_i = F(\psi_i).$$

Then

$$\boxed{\mathbf{KU} = \mathbf{F}.} \quad (6)$$

The unknown function has therefore been replaced by  $N$  unknown coefficients.

### Why an arbitrary Galerkin basis is not yet a finite element basis

The basis functions  $\psi_i$  could be global functions such as polynomials or trigonometric functions. Such functions are generally nonzero over most or all of the domain. Consequently  $B(\psi_j, \psi_i)$  is usually nonzero for many pairs  $(i, j)$  and  $\mathbf{K}$  is dense.

The finite element method keeps the Galerkin idea but constructs basis functions that are nonzero only on small parts of the domain. This *local support* is the key structural difference.

### A simple global Galerkin example

For a uniform axial bar with constant  $AE$ , no distributed load,  $u(0) = 0$ , and end load  $P$ , the exact solution is

$$u(x) = \frac{P}{AE}x.$$

Take  $\psi_1(x) = 1$  and  $\psi_2(x) = x$ . Then

$$u_N = U_1 + U_2x.$$

The essential condition gives  $U_1 = 0$ . The Galerkin equation then yields

$$AE U_2 L = PL, \quad U_2 = \frac{P}{AE},$$

so this space happens to contain the exact solution. The example is useful, but these basis functions are global. FEM will instead construct basis functions from a mesh.

### 3 Mesh, elements, nodes, and the finite element basis

#### 3.1 Partition of the interval

Let

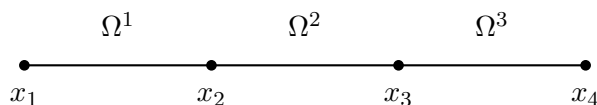
$$0 = x_1 < x_2 < \cdots < x_N = L$$

be global nodes. The domain is partitioned into elements

$$\Omega^e = [x_1^e, x_2^e], \quad e = 1, \dots, N_e,$$

where neighboring elements share endpoint nodes. For a linear three-element mesh,

$$\Omega^1 = [x_1, x_2], \quad \Omega^2 = [x_2, x_3], \quad \Omega^3 = [x_3, x_4].$$



The element length is

$$h^e = x_2^e - x_1^e.$$

The collection of all elements is the mesh  $\mathcal{T}_h$ .

#### 3.2 Three requirements for nodal finite element basis functions

The lecture construction can be summarized by three requirements.

1. **Polynomial on each element.** On an element  $\Omega^e$ , the restriction of a basis function is a polynomial of prescribed degree  $p^e$ .
2. **Continuous across element boundaries.** For the  $H^1$ -conforming finite elements considered here, neighboring polynomial pieces join continuously at shared nodes.
3. **Nodal interpolation property.** The global basis function  $\phi_i$  associated with global node  $x_i$  satisfies

$$\phi_i(x_j) = \delta_{ij}. \tag{7}$$

These conditions create basis functions tied directly to the mesh rather than to the whole domain.

#### 3.3 Begin with one two-node linear element

Consider one element

$$\Omega^e = [x_1^e, x_2^e].$$

A linear element has two local nodes and therefore two local basis functions. Let

$$\phi_1^e(x) = a_1 + b_1x, \quad \phi_2^e(x) = a_2 + b_2x.$$

The nodal conditions are

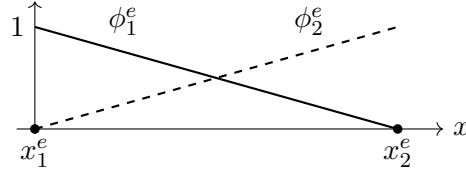
$$\begin{aligned} \phi_1^e(x_1^e) &= 1, & \phi_1^e(x_2^e) &= 0, \\ \phi_2^e(x_1^e) &= 0, & \phi_2^e(x_2^e) &= 1. \end{aligned}$$

Solving these two small interpolation problems gives

$$\boxed{\phi_1^e(x) = \frac{x_2^e - x}{h^e}, \quad \phi_2^e(x) = \frac{x - x_1^e}{h^e}.} \quad (8)$$

Thus

$$\phi_a^e(x_b^e) = \delta_{ab}, \quad a, b = 1, 2.$$



The two functions form a local basis on the element. A finite element function restricted to this element has the form

$$u_h|_{\Omega^e} = U_1^e \phi_1^e + U_2^e \phi_2^e. \quad (9)$$

Because of the nodal property,

$$u_h(x_1^e) = U_1^e, \quad u_h(x_2^e) = U_2^e.$$

### 3.4 Patching local linear functions into global basis functions

Now return to the three-element, four-node mesh. Each global basis function is constructed by combining the appropriate local linear pieces and setting the function to zero elsewhere.

For example,

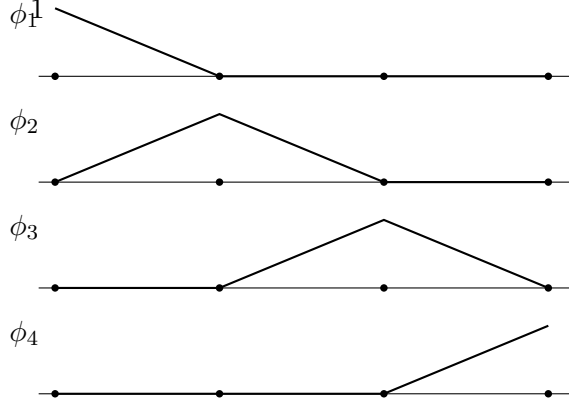
$$\phi_1(x) = \begin{cases} \phi_1^1(x), & x \in \Omega^1, \\ 0, & \text{otherwise,} \end{cases}$$

while the basis function associated with the shared node  $x_2$  is

$$\phi_2(x) = \begin{cases} \phi_2^1(x), & x \in \Omega^1, \\ \phi_1^2(x), & x \in \Omega^2, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly,

$$\phi_3(x) = \begin{cases} \phi_2^2(x), & x \in \Omega^2, \\ \phi_1^3(x), & x \in \Omega^3, \\ 0, & \text{otherwise,} \end{cases} \quad \phi_4(x) = \begin{cases} \phi_2^3(x), & x \in \Omega^3, \\ 0, & \text{otherwise.} \end{cases}$$



The diagrams make two important properties visible:

- $\phi_i(x_j) = \delta_{ij}$ ;
- each  $\phi_i$  is nonzero only on the element or elements touching node  $x_i$ .

The second property is the local support responsible for sparsity.

The global linear finite element space is

$$\mathcal{V}_h = \{v_h \in C([0, L]) : v_h|_{\Omega^e} \in \mathbb{P}_1(\Omega^e) \text{ for each } e\}.$$

Every  $u_h \in \mathcal{V}_h$  can be written

$$u_h(x) = \sum_{i=1}^N U_i \phi_i(x). \quad (10)$$

Using (7),

$$u_h(x_j) = \sum_i U_i \phi_i(x_j) = U_j.$$

Therefore the coefficients are the nodal values of the finite element function.

## 4 Higher-order elements

The same construction extends naturally when each element contains more than two nodes.

### 4.1 A three-node quadratic element

Consider one element with local nodes

$$x_1^e < x_2^e < x_3^e.$$

A quadratic element has three local basis functions. Each is a polynomial of degree at most two and satisfies

$$\phi_a^e(x_b^e) = \delta_{ab}, \quad a, b = 1, 2, 3.$$

For example, the first function is the unique quadratic satisfying

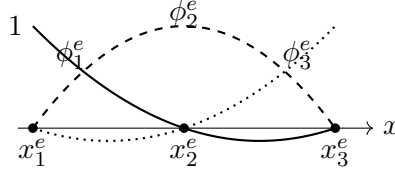
$$\phi_1^e(x_1^e) = 1, \quad \phi_1^e(x_2^e) = 0, \quad \phi_1^e(x_3^e) = 0.$$

The three functions can be written directly using Lagrange interpolation:

$$\phi_1^e(x) = \frac{(x - x_2^e)(x - x_3^e)}{(x_1^e - x_2^e)(x_1^e - x_3^e)}, \quad (11)$$

$$\phi_2^e(x) = \frac{(x - x_1^e)(x - x_3^e)}{(x_2^e - x_1^e)(x_2^e - x_3^e)}, \quad (12)$$

$$\phi_3^e(x) = \frac{(x - x_1^e)(x - x_2^e)}{(x_3^e - x_1^e)(x_3^e - x_2^e)}. \quad (13)$$



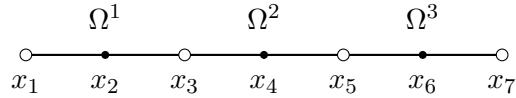
The endpoint basis functions are not simply straight lines anymore. They may also take negative values inside the element. What remains essential is the nodal interpolation property and polynomial degree.

## 4.2 Three quadratic elements and seven global nodes

For three quadratic elements, neighboring elements share only their endpoint nodes. A typical uniform mesh has

$$\Omega^1 = [x_1, x_3], \quad \Omega^2 = [x_3, x_5], \quad \Omega^3 = [x_5, x_7],$$

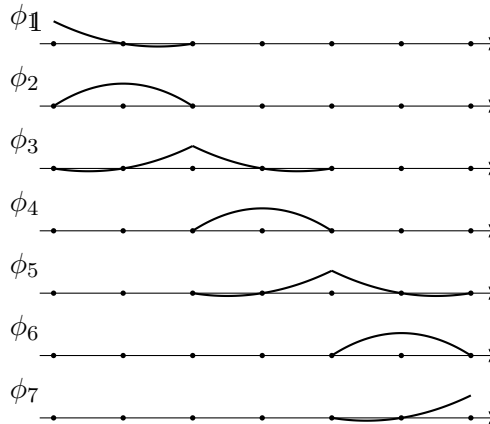
with midpoint nodes  $x_2, x_4, x_6$ .



The local-to-global node connectivity is

$$I^1 = (1, 2, 3), \quad I^2 = (3, 4, 5), \quad I^3 = (5, 6, 7).$$

The seven global basis functions are again obtained by patching local quadratic pieces. The midpoint functions are supported on one element; basis functions associated with shared element endpoints are supported on two neighboring elements.



This picture is the higher-order analogue of the linear hat-function picture. The shapes differ, but the construction rules are unchanged.

### 4.3 General polynomial degree

A degree- $p$  one-dimensional Lagrange element has  $p + 1$  local nodes and  $p + 1$  local basis functions. On each element the basis functions are polynomials of degree at most  $p$ , they join continuously at shared endpoint nodes, and they satisfy the nodal property. The global finite element space is

$$\mathcal{V}_h^p = \{v_h \in C([0, L]) : v_h|_{\Omega^e} \in \mathbb{P}_p(\Omega^e) \text{ for every } e\}. \quad (14)$$

## 5 The reference or master element

Constructing new polynomial formulas separately on every physical element is unnecessary. Every one-dimensional element can be related to one fixed reference element

$$\widehat{\Omega} = [-1, 1],$$

with coordinate  $\xi$ .

### 5.1 Coordinate mapping

For the physical element

$$\Omega^e = [x_1^e, x_2^e], \quad h^e = x_2^e - x_1^e,$$

seek a linear map  $x^e(\xi) = m\xi + c$  such that

$$x^e(-1) = x_1^e, \quad x^e(1) = x_2^e.$$

Solving gives

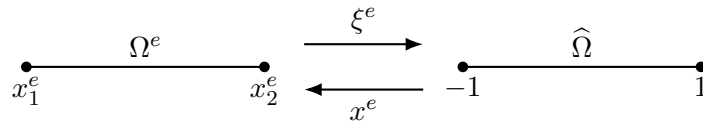
$$\boxed{x^e(\xi) = \frac{x_1^e + x_2^e}{2} + \frac{h^e}{2}\xi.} \quad (15)$$

The inverse map is

$$\boxed{\xi^e(x) = \frac{2}{h^e} \left( x - \frac{x_1^e + x_2^e}{2} \right).} \quad (16)$$

The Jacobians are

$$J_x^e = \frac{dx^e}{d\xi} = \frac{h^e}{2}, \quad J_\xi^e = \frac{d\xi^e}{dx} = \frac{2}{h^e} = \frac{1}{J_x^e}. \quad (17)$$



## 5.2 Lagrange basis functions on the master element

For degree  $p$ , choose  $p + 1$  distinct master-element nodes

$$\xi_1, \dots, \xi_{p+1}.$$

The local reference basis function associated with node  $i$  is

$$\hat{\phi}_i(\xi) = \prod_{\substack{k=1 \\ k \neq i}}^{p+1} \frac{\xi - \xi_k}{\xi_i - \xi_k}. \quad (18)$$

At a reference node  $\xi_j$ , every factor is finite and the numerator contains  $\xi_j - \xi_j = 0$  unless  $i = j$ . Hence

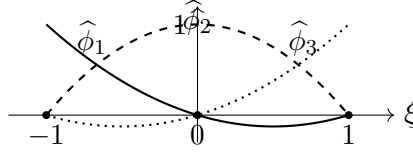
$$\hat{\phi}_i(\xi_j) = \delta_{ij}. \quad (19)$$

For  $p = 1$ , with nodes  $-1$  and  $1$ ,

$$\hat{\phi}_1(\xi) = \frac{1 - \xi}{2}, \quad \hat{\phi}_2(\xi) = \frac{1 + \xi}{2}.$$

For  $p = 2$ , with nodes  $-1, 0, 1$ ,

$$\hat{\phi}_1(\xi) = \frac{1}{2}\xi(\xi - 1), \quad \hat{\phi}_2(\xi) = 1 - \xi^2, \quad \hat{\phi}_3(\xi) = \frac{1}{2}\xi(\xi + 1). \quad (20)$$



The physical-element basis functions are obtained by composition:

$$\phi_i^e(x) = \hat{\phi}_i(\xi^e(x)). \quad (21)$$

The derivative follows from the chain rule:

$$\frac{d\phi_i^e}{dx} = \frac{d\hat{\phi}_i}{d\xi} \frac{d\xi^e}{dx} = J_\xi^e \frac{d\hat{\phi}_i}{d\xi}. \quad (22)$$

## 5.3 Local and global node numbering

The element formulas use local numbers  $1, \dots, p + 1$ , while the mesh uses global node numbers. A connectivity table provides the correspondence.

For three linear elements,

| element | local node 1 | local node 2 |
|---------|--------------|--------------|
| 1       | 1            | 2            |
| 2       | 2            | 3            |
| 3       | 3            | 4            |

For three quadratic elements,

| element | local node 1 | local node 2 | local node 3 |
|---------|--------------|--------------|--------------|
| 1       | 1            | 2            | 3            |
| 2       | 3            | 4            | 5            |
| 3       | 5            | 6            | 7            |

If  $I = I^e(a)$  denotes the global node corresponding to local node  $a$  of element  $e$ , then on that element

$$\phi_I|_{\Omega^e} = \phi_a^e. \quad (23)$$

## 6 The finite element variational problem

Let  $\mathcal{V}_{h,0} \subset \mathcal{V}$  be the finite element space satisfying the homogeneous essential boundary condition. The finite element problem is

$$\boxed{\text{find } u_h \in \mathcal{V}_{h,0} \text{ such that } B(u_h, v_h) = F(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.} \quad (24)$$

Write

$$u_h = \sum_{j=1}^N U_j \phi_j, \quad v_h = \sum_{i=1}^N V_i \phi_i.$$

Exactly as in the Galerkin derivation,

$$\sum_{j=1}^N B(\phi_j, \phi_i) U_j = F(\phi_i), \quad i = 1, \dots, N.$$

Thus

$$K_{ij} = B(\phi_j, \phi_i), \quad F_i = F(\phi_i), \quad \boxed{\mathbf{KU} = \mathbf{F}.}$$

The important difference from a generic global Galerkin basis is that each  $\phi_i$  has local support. Therefore most pairs of basis functions do not overlap and most  $K_{ij}$  vanish.

## 7 Element matrices and element load vectors

Because a global basis function is nonzero only on a few elements, every global integral can be decomposed into element integrals. For an element  $\Omega^e$  with local basis functions  $\phi_a^e$ , define

$$K_{ab}^e = \int_{\Omega^e} \left[ a(x) \frac{d\phi_b^e}{dx} \frac{d\phi_a^e}{dx} + c(x) \phi_b^e \phi_a^e \right] dx, \quad (25)$$

$$F_a^e = \int_{\Omega^e} f(x) \phi_a^e(x) dx. \quad (26)$$

These are the local element stiffness matrix and load vector. Natural boundary terms are added to the appropriate global load entry.

## Two-node linear bar element

For  $a = AE$ ,  $c = 0$ , and a two-node linear element,

$$\frac{d\phi_1^e}{dx} = -\frac{1}{h^e}, \quad \frac{d\phi_2^e}{dx} = \frac{1}{h^e}.$$

Hence

$$\mathbf{K}^e = \frac{A_e E_e}{h^e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \quad (27)$$

If  $f$  is constant on the element,

$$\mathbf{F}^e = \frac{f_e h^e}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (28)$$

## 8 Local-to-global assembly in detail

Assembly is the process of adding the local element contributions into the correct global rows and columns. The connectivity table determines where each local entry belongs.

### 8.1 Assembly of the load vector

For the three-linear-element mesh with connectivity

$$I^1 = (1, 2), \quad I^2 = (2, 3), \quad I^3 = (3, 4),$$

write

$$\mathbf{F}^e = \begin{bmatrix} F_1^e \\ F_2^e \end{bmatrix}.$$

Then

$$\begin{aligned} F_1 &= F_1^1, \\ F_2 &= F_2^1 + F_1^2, \\ F_3 &= F_2^2 + F_1^3, \\ F_4 &= F_2^3 + q_L. \end{aligned}$$

The shared nodes receive contributions from both neighboring elements.

### 8.2 Assembly of the stiffness matrix

Let

$$\mathbf{K}^e = \begin{bmatrix} K_{11}^e & K_{12}^e \\ K_{21}^e & K_{22}^e \end{bmatrix}.$$

The assembled matrix is

$$\mathbf{K} = \begin{bmatrix} K_{11}^1 & K_{12}^1 & 0 & 0 \\ K_{21}^1 & K_{22}^1 + K_{11}^2 & K_{12}^2 & 0 \\ 0 & K_{21}^2 & K_{22}^2 + K_{11}^3 & K_{12}^3 \\ 0 & 0 & K_{21}^3 & K_{22}^3 \end{bmatrix}. \quad (29)$$

This makes the local-to-global rule explicit. In general, if local nodes  $a, b$  of element  $e$  correspond to global nodes  $I, J$ , then

$$F_I += F_a^e, \quad K_{IJ} += K_{ab}^e. \quad (30)$$

### 8.3 Essential boundary conditions

If  $x_1 = 0$  is a prescribed zero-displacement node, then the nodal coefficient  $U_1 = u_h(x_1)$  is known to be zero. The test coefficient associated with that node is also zero. The algebraic equations are therefore solved only for the free coefficients. For nonzero prescribed nodal values, the known values are moved to the right-hand side in the usual partitioned matrix system.

## 9 A complete two-element bar calculation

Consider a uniform bar with constant  $AE$ , length  $L$ , no distributed load,  $u(0) = 0$ , and end load  $P$  at  $x = L$ . Divide the bar into two equal elements,  $h = L/2$ . Each contributes

$$\mathbf{K}^e = \frac{AE}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Assembly gives

$$\frac{AE}{h} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ P \end{bmatrix}.$$

Using  $U_1 = 0$  gives

$$\frac{AE}{h} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} 0 \\ P \end{bmatrix}.$$

Therefore

$$U_2 = \frac{PL}{2AE}, \quad U_3 = \frac{PL}{AE}.$$

Since the exact solution  $u(x) = Px/(AE)$  is linear, the linear finite element space reproduces it exactly.

## 10 Transforming element integrals to the master element

Let  $g$  be integrable on  $\Omega^e$ . Using  $x = x^e(\xi)$  and  $dx = J_x^e d\xi$ ,

$$\boxed{\int_{\Omega^e} g(x) dx = J_x^e \int_{-1}^1 g(x^e(\xi)) d\xi.} \quad (31)$$

This is the reason the reference element is so useful: every element integral has the same fixed limits  $[-1, 1]$ .

For the load vector,

$$F_a^e = J_x^e \int_{-1}^1 f(x^e(\xi)) \widehat{\phi}_a(\xi) d\xi. \quad (32)$$

For the derivative part of the element matrix,

$$\int_{\Omega^e} a \frac{d\phi_b^e}{dx} \frac{d\phi_a^e}{dx} dx = \frac{1}{J_x^e} \int_{-1}^1 a(x^e(\xi)) \widehat{\phi}_b'(\xi) \widehat{\phi}_a'(\xi) d\xi. \quad (33)$$

The *cuv* contribution becomes

$$\int_{\Omega^e} c \phi_b^e \phi_a^e dx = J_x^e \int_{-1}^1 c(x^e(\xi)) \widehat{\phi}_b(\xi) \widehat{\phi}_a(\xi) d\xi. \quad (34)$$

## 11 Gauss–Legendre quadrature

The reference-element integrals need not always be evaluated symbolically. Gauss–Legendre quadrature approximates

$$\int_{-1}^1 g(\xi) d\xi \approx \sum_{q=1}^Q w_q g(\xi_q), \quad (35)$$

where  $\xi_q$  are quadrature points and  $w_q$  are weights.

| $Q$ | quadrature points   | weights       | exact for polynomials through degree |
|-----|---------------------|---------------|--------------------------------------|
| 1   | 0                   | 2             | 1                                    |
| 2   | $\pm 1/\sqrt{3}$    | 1, 1          | 3                                    |
| 3   | $0, \pm \sqrt{3/5}$ | 8/9, 5/9, 5/9 | 5                                    |

Therefore

$$F_a^e \approx J_x^e \sum_{q=1}^Q w_q f(x^e(\xi_q)) \hat{\phi}_a(\xi_q), \quad (36)$$

$$K_{ab}^e \approx \sum_{q=1}^Q w_q \left[ \frac{a(x^e(\xi_q))}{J_x^e} \hat{\phi}_b'(\xi_q) \hat{\phi}_a'(\xi_q) + J_x^e c(x^e(\xi_q)) \hat{\phi}_b(\xi_q) \hat{\phi}_a(\xi_q) \right]. \quad (37)$$

For constant  $AE$ , the linear-element stiffness integrand is constant, so one Gauss point is sufficient. For a quadratic element with constant  $AE$ , the stiffness integrand is quadratic, so two Gauss points integrate it exactly.

## 12 Summary of the one-dimensional FEM procedure

The full procedure can now be written as an element-by-element algorithm.

1. Start from the variational problem  $B(u, v) = F(v)$ .
2. Create the mesh and the element-to-global connectivity table.
3. Choose the polynomial degree  $p$  and the local nodal layout.
4. Define the local Lagrange basis functions on the master element.
5. For each element, construct the coordinate map  $x^e(\xi)$  and its Jacobian.
6. Evaluate the local element matrix  $\mathbf{K}^e$  and load vector  $\mathbf{F}^e$ , usually by Gauss quadrature.
7. Assemble local entries into the global matrix and vector using the connectivity table.
8. Add natural boundary contributions to the global load vector.
9. Impose the prescribed nodal values and solve the reduced algebraic system.
10. Reconstruct  $u_h(x)$  and compute derived quantities such as strain, stress, or flux.

The central ideas to retain are not the individual formulas but the structure: local polynomial approximation, nodal basis functions, reference-to-physical mapping, local element calculations, and global assembly. These ideas remain unchanged when the method is extended to higher dimensions.